higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

MATHEMATICAL LITERACY
(First Paper)
NQF LEVEL 4

5 November 2019 (X-Paper)
09:00–12:00

This marking guideline consists of 10 pages.

Symbol	Explanation
M	Method
MA	Method with accuracy
CA	Consistent accuracy
A	Accuracy
C	Conversion
S	Simplification
RT/RG/RD	Reading from a table/Reading from a graph/drawing
F	Choosing correct formula
SF	Substitution in formula
R/J	Reasoning/Justification
P	Penalty, e.g. for no units, incorrect rounding off etc.
R	Rounding off
MF	Manipulating Formula

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QUESTION 1 [30] *Do NOT deduct marks if the 'R' or the '%' sign are omitted.		
Question	**Solution**	**Explanation**
1.1	$\frac{7^2-\sqrt{36}+\sqrt[3]{8}}{6+4(19+2)}$ $=\frac{49-6+2}{6+84}$ ✓ $=\frac{45}{90}$ ✓ $=\frac{1}{2}$ ✓	1 numerator 1 denominator 1 S 1A (4)
1.2	$\frac{1}{2};\frac{3}{5};\frac{1}{3}$ $=\frac{15}{30};\frac{18}{30};\frac{10}{30}$ ✓ $=\frac{1}{3}$✓;$\frac{1}{2}$✓;$\frac{3}{5}$✓	1 M LCD 3A - correct ascending order (4)
1.3	$23,2\times1000$ ✓ = 23 200 ✓ ℓ $\therefore 23\,200\times1\,000$ = ✓23 200 000 ✓ mℓ Or $23,2\times1\,000\,000$ ✓✓✓ = 23 200 000✓ mℓ	1 M 1 C 23 200 1 C × 1 000 1 A 2 320 000 mℓ 2 C 1 000 000 1 M $23,2\times1\,000\,000$ 1 A 2 320 000 mℓ (4)
1.4	Rise in temperature = 23 °C - (-2 °C)✓ = 25°C✓ (answer only full marks)	1 M 1 A (2)
1.5	Monday 6:53 pm to Wednesday 6:53 pm = 36 hours✓ Wednesday 6:53 to 11:02 am = 4 hours and 9min✓ 11:02 - 6:53 4:09 Total time passed = 40 hours 9 min✓	1 MA 1 A 1 A 1 M 1 A (3)

1.6	$\text{Price of shoe} = \dfrac{R14\,105\checkmark \times 100\checkmark}{105\checkmark}$ $= \text{R13 433,33}\checkmark$	3 MA 1 A (4)
1.7	$\text{R1 300} = \dfrac{\text{R1 300}}{\text{£17,52}}\checkmark$ $= \text{£ 74,2}\checkmark$ $= \text{£ 74}\checkmark$	1 M 1 A 1 R (3 answer only) (3)
1.8	1 kg = 250 g × 4 $\therefore \frac{1}{4} kg\ tin = \dfrac{\text{R99,99}}{4\checkmark} = \underline{\text{R24,99 per 250 grams}}\checkmark$ but a 250-gram tin = <u>R33,99</u> Therefore the 1 kg tin is more cost effective✓ Or 50 g × 4 = R33,99 × 4✓ 1 kg = R135,96✓ But 1 kg tin = R99,99 Therefore the 1 kg tin is more cost effective✓	1 M 1 A 1 E 1 M 1 A 1 E (3)
1.9	$\text{Tumeric}: \frac{3}{9} \times 162 = 54g$ ✓ $\text{Cayenne}: \frac{2}{9} \times 162 = 36g$ ✓ $\text{Cinnamon}: \frac{4}{9} \times 162 = 72g$ ✓ :	1 A 1A 1A (3)

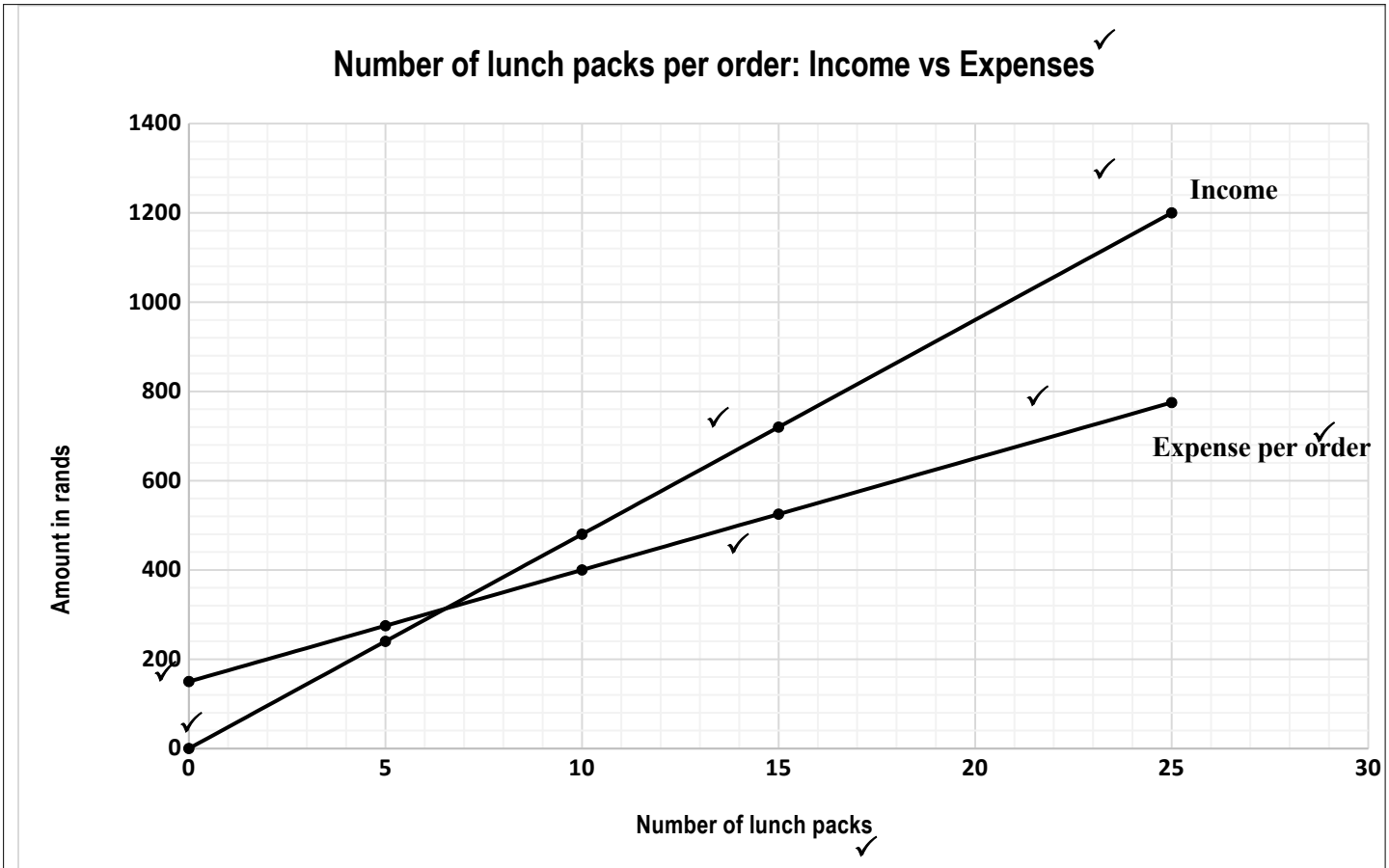
QUESTION 2 [30] *Penalise once for incorrect unit unless stated differently			
Question		**Solution**	**Explanation**
2.1	2.1.1	Table Mountain✓✓ Kirstenbosch Gardens✓✓, Chapmans Peak✓✓ Any other tourist attraction	2 RM (2)
	2.1.2	23 refreshment stations✓✓	2 RM (2)
	2.1.3	Chapman's Peak drive✓✓	2 RM (2)
	2.1.4	12 km✓✓	2 RM (2)
	2.1.5	Approximate distance ≈ 34 km✓ – 18 km✓ ≈ 16 km✓	2 RM and M 1 A (3)

	2.1.6	1 : 500 000 33,6 mm : Actual distance in mm $\therefore$ Actual distance in mm = 33,6✓ × 500 000✓ = 16 800 000 mm ✓ Actual distance in km = 16 800 000 mm ÷ 1000 000✓ = 16,8 km ✓	2 M 1 A 1 C 1 CA (5)
2.2	2.2.1	$AC^2 = AD^2 + DC^2$ $40^2 = h^2 + 20^2$✓ $h^2 = 40^2$✓-20^2✓ $h^2 = 1200$✓ $h = \sqrt{1200}$✓ $h = 34,64\,mm$ OR $AC^2 = AD^2 + DC^2$ $h^2 = AD^2$✓$-DC^2$✓ $h = \sqrt{AD^2 - DC^2}$✓ $h = \sqrt{40^2 - 20^2}$✓ $h = \sqrt{1200}$✓ $h = 34,64\,mm$	1 SF 2 MF 1 A 1 MA 2 MF 1 MF 1 SF 1 A (5)
	2.2.2	$\text{Area} = \frac{1}{2} \times \text{base} \times \text{height}$ $= \frac{1}{2} \times 40$✓$\times 34,64$✓ $= 692,8$✓ mm^2✓ (If 20 is substituted then CA **346** with 3 marks, if correct unit. Otherwise 2 marks))	2 SF 1 A and 1 unit (4)
	2.2.3	Total surface area $= 2 \times$ area of triangular face $+ 3 \times (l \times w) +$ area of flaps $=2 \times 692,8$✓$+3(210$✓$\times 40$✓$) + 370$✓ $= 26\,955,6$✓ mm^2	4 SF 1 CA (Q2.2.2) (5)

QUESTION 3 [30] * (Do NOT penalise if 'r' sign is omitted)			
Question		**Solution**	**Explanation**
3.1	3.1.1	F	1 A (1)
	3.1.2	C	1 A (1)
	3.1.3	A	1 A (1)
	3.1.4	E	1 A (1)
	3.1.5	D	1 A (1)
3.2	3.2.1	Minimum gross monthly income for 15 years = R30 447,15✓ Anele's monthly gross income = $\frac{331\,000}{12}$ ✓ =R27 583,33✓ No✓, she does not earn the minimum gross salary	1 RT 1 M 1A 1A (4)
	3.2.2	20 years × 12 months = 240✓ Payment over 20 years = R8 202,68 × 240✓ = R1 968 643,20✓ Total interest = R1 968 643,20 – R850 000✓ = R1 118 643,20✓	1 MA 1 M 1 A 1 M 1 CA (5)
	3.2.3	Service fees over 20 years = R69✓ × 240✓ = R16 560✓ (Answer only full marks)	1 RT and 1 M 1 A (3)
	3.2.4	Total cost = R1 968 643,20✓ + R16 560✓ = R1 985 203,20	2 MA (2)
	3.2.5	Total saving = R2 710 209,60✓ – R1 985 203,20✓ = R725 006,43✓	1 RT and 1 M 1 A (3)
3.3	3.3.1	Housing subsidy = R27 583,33 – R26 083,33✓ = R1 500✓ **(Answer only full marks)**	1 M 1 A (2)

	3.3.2	UIF = 0,01✓ × R27 583,33 = R275,83✓ PAYE = 0,22 × R27 583,33 =R6 068,33✓ Total deductions = R8 058,60✓ Net salary = R27 583,33 - R8 058,60✓ = R19 524,73✓	1 M and 1 A 1 A 1 CA 1 M 1 CA (6)

QUESTION 4 [30] * (Do NOT penalise if 'r' sign is omitted)		
Question	**Solution**	**Explanation**
4.1	A = R400✓✓ B = R720✓✓	1 RT 1 A 1 RT 1 A (4)
4.2	Constant difference. ✓ R48✓ (48; 96; 144….) Accept R240 (240; 480; 720….)	1 A constant difference 1 A R48 (2)
4.3	Expenses per order in rands = R150✓+ R25✓ × Number of lunch packs✓ Or Expenses per order in rands = 150✓ + 25✓*n*✓ (Accept other correct alternatives)	1A – R150 1 A – R25 1 A - × No. of lunch packs (3)
4.4	Income per order in rands = R48✓ × Number of lunch packs✓ Or Income per order in rands = 48✓ × *n*✓ (Accept other correct alternatives)	1 A – R48 1 A - × Number of lunch packs (2)
4.5	See graph below	1 labelling title 1 labelling axes (no mark if one is omitted) 1 labelling both lines (no mark if one is omitted) 3 line graph –expense per order: (2 for plotting and 1 for line) 3 line graph – Income: (2 for plotting and 1 for line) (9)



4.6	4.6.1	Direct proportion. As the number of lunch packs increases✓the income increases in the same proportion.✓	2 E (2)
	4.6.2	Profit = (660 to 680) ✓ – (500 to 520)✓ = R160 approximately✓ (NO marks for R672 – R500 = R172)	1 RG and 1 M 1 CA (3)
	4.6.3	Income – expenses = R240 – R275✓ = - R35 A loss✓ of R35✓	1 M 1 A Loss 1 A R35 (3)
	4.6.4	7✓✓	2 RG (2)

QUESTION 5 [30] * (Do NOT penalise if 'r' sign is omitted)

Question		Solution	Explanation
5.1	5.1.1	Housebreaking or burglary✓✓	2 RT (2)
	5.1.2	Theft of bicycle✓✓	2 RT (2)
	5.1.3	Range = 777 000✓ – 15 000✓ = 762 0000✓	2 M/RT 1 A (3)
	5.1.4	Percentage decrease = $\frac{160\ ✓ - 151✓}{160✓} \times 100✓$ = 5,625% ✓ Accept -5,625%	4 MA 1 A (5)
	5.1.5	Average 2015/2016 = $\frac{1\,258\,000✓}{10✓}$ = 125 800✓\ Average 2016/2017 = $\frac{1\,407\,000}{10}$ = 140 700✓ Average increase = 140 700 – 125 800✓ = 14 900✓	2 M 1 A 1 A 1 M 1 CA (6)
	5.1.6	Ascending order: Theft of crops planted by the household 5 000✓ Murder 16 000 Theft of bicycle 21 000 Motor vehicle vandalism 32 000 Deliberate damaging, burning, destruction of buildings 47 000 Theft of motor vehicle 48 000 Theft out of motor vehicle 139 000 Home robbery 151 000 Theft of livestock, poultry and other animals 161 000 Housebreaking or burglary 777 000✓ Median = $\frac{47+48}{2}$ = **47,5**✓ Deliberate damaging, burning, destruction of buildings and Theft of motor vehicle✓	2 A – ascending order 1 M - locating the position of the median 1 A – median (4)

5.2	5.2.1	• The scale of the vertical axis is different.✓ • The scale of the horizontal axis is different. • The vertical axis of graph A begins at 20 but the vertical axis of graph B starts 0.	3 R/J (3)
	5.2.2	Strong relationship✓	1A (1)
	5.2.3	Representative 2✓	1 RT/RG (1)
	5.2.4	Reskilling✓ OR Closely monitor phone calls made by representative 2✓ OR Any other appropriate answer	1 E (1)
	5.2.5	Graph B✓ The scale helps with meaningful reading✓ OR Any other appropriate answer	1 A 1R/J (2)
		TOTAL	**150**

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